

ALGEBRA – UNIT 1

Polytechnic Engineering Mathematics - Easy Hindi Notes

PDF Polytechnic Algebra topics Unit-1 topics

1. Algebra ☐☐☐☐ ☐☐

Algebra **Variables** **Equations**

variables x, y, a, b expressions equations: $2x + 5 = 15$.

2. Algebraic Expressions

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Term	Algebraic Expression	Meaning
Constant	5	Fixed value, does not change
Variable	x	Value that can change or vary
Coefficient	3x or 3	Number multiplying the variable
Expression	2x + 3	Combination of terms

3. Laws of Indices (指数的法則)

- $a^m \times a^n = a^{m+n}$
- $a^m \div a^n = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $(ab)^n = a^n b^n$
- $a^0 = 1$ ($a \neq 0$)
- $a^{-n} = 1/a^n$

□□□□□: $2^3 \times 2^2 = 2^5 = 32$.

4. Algebraic Identities

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $a^2 - b^2 = (a - b)(a + b)$
- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

5. Factorisation

Factorisation 因式分解 algebraic expression 代数表达式 factors 因数 质因数 分解 质因数分解

- $x^2 + 5x + 6 = (x + 2)(x + 3)$
- $x^2 - 9 = (x - 3)(x + 3)$
- $4x^2 - 12x = 4x(x - 3)$

6. Linear Equation

一次 equation 一次 variable 变量 highest power 1 次, 一次 linear equation 一次方程 线性方程

例题: $3x + 5 = 20$

解: $3x = 20 - 5 = 15 \Rightarrow x = 5$

7. Quadratic Equation — 重要公式

quadratic equation 二次方程式 $ax^2 + bx + c = 0$ 中, $a \neq 0$, 二次方程 二次方程

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Discriminant: $D = b^2 - 4ac$

$D > 0 \rightarrow$ 两个实根; $D = 0 \rightarrow$ 一个实根; $D < 0 \rightarrow$ 没有实根

8. Sequence & Series — Basic

Arithmetic Progression (AP) 等差数列 consecutive terms 连续项 difference 公差

AP 第 n 项: $a_n = a + (n - 1)d$

n 项和: $S_n = \frac{n}{2} [2a + (n - 1)d]$

例子: 2, 5, 8, 11, ... $a = 2$ $d = 3$.

9. Important Questions

- Algebraic expression, term, coefficient 和 constant 的识别
- Indices 的 laws 的应用
- Algebraic identities 的应用
- $x^2 + 7x + 12$ 的因式分解
- $4x^2 - 25$ 的因式分解
- $3x + 7 = 25$ 的求解
- Quadratic equation 的标准形式 和 quadratic formula 的应用
- Discriminant 的应用
- AP 第 n 项 和 sum 的公式

10. Quick Revision Formula Sheet

Topic	Formula
Indices	$a^m \times a^n = a^{m+n}$
Square	$(a+b)^2 = a^2+2ab+b^2$
Difference	$a^2-b^2=(a-b)(a+b)$
Cubes	$a^3 \pm b^3$ formulas
Quadratic	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Discriminant	$D = b^2 - 4ac$
AP nth term	$a_n = a + (n - 1)d$
AP sum	$S_n = \frac{n}{2} [2a + (n - 1)d]$

Tip: 4 formulas 400 4000, 400 40 formula 40 40-40-40 3 numerical questions practice 40000